OS & SP

Name : S LOHITHA **Roll No :** 2520030109

SKILL – 23

1.Skill23.c

lohitha27@LAPTOP-A9F545F(

GNU nano 8.7.1 skill23.c *

```c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <sys/types.h>
#include <time.h>
#define JOBS 4
int main()
{
    pid_t pids[JOBS];

    printf("====================================\n");
    printf(" OSSP PROGRAM 23\n");
    printf(" Large Pipeline & Performance Test\n");
    printf("====================================\n\n");

    printf("Launching %d jobs...\n\n", JOBS);
    fflush(stdout);

    /* Launch multiple jobs */
    for (int i = 0; i < JOBS; i++) {

        pids[i] = fork();

        if (pids[i] < 0) {
            perror("fork");
            return 1;
        }

        /* Child process */
        if (pids[i] == 0) {

            printf("Job %d started | PID = %d\n",
                   i + 1, getpid());
            fflush(stdout);

            /* Simulate different workloads */
            sleep(i + 1);

            printf("Job %d completed | PID = %d\n",
                   i + 1, getpid());
            fflush(stdout);

            exit(0);
        }
    }
    /* Parent waits for all jobs */
    printf("\nParent waiting for all jobs...\n\n");
    for (int i = 0; i < JOBS; i++) {
        int status;
        if (waitpid(pids[i], &status, 0) < 0) {
            perror("waitpid");
            continue;
        }
        if (WIFEXITED(status)) {

            printf("Job %d finished successfully "
                   "with status %d\n",
                   i + 1, WEXITSTATUS(status));

        } else if (WIFSIGNALED(status)) {

            printf("Job %d failed with signal %d\n",
                   i + 1, WTERMSIG(status));
        }
    }
    printf("\n====================================\n");
    printf("All jobs have been processed.\n");
    printf("Performance test completed.\n");
    printf("====================================\n");
    return 0;
}
```

OUTPUT :

```
lohitha27@LAPTOP-A9F545FG:~$ gcc skill23.c -o skill23
lohitha27@LAPTOP-A9F545FG:~$ ./skill23
====================================
 OSSP PROGRAM 23
 Large Pipeline & Performance Test
====================================

Launching 4 jobs...

Job 1 started | PID = 604
Job 2 started | PID = 605
Job 3 started | PID = 606

Parent waiting for all jobs...

Job 4 started | PID = 607
Job 1 completed | PID = 604
Job 1 finished successfully with status 0
Job 2 completed | PID = 605
Job 2 finished successfully with status 0
Job 3 completed | PID = 606
Job 3 finished successfully with status 0
Job 4 completed | PID = 607
Job 4 finished successfully with status 0

====================================
All jobs have been processed.
Performance test completed.
====================================
lohitha27@LAPTOP-A9F545FG:~$
```